

Project Title

Teleoperated Surgical Robotic Arm with Real-Time Vision Feedback

1. Introduction

Robotic-assisted surgery is one of the most significant technological advancements in modern healthcare. Teleoperated robotic systems allow surgeons to control robotic instruments remotely while receiving real-time visual feedback from the surgical environment. Such systems can improve precision, reduce human error, and potentially provide medical expertise to remote locations.

This project aims to develop a low-cost prototype of a teleoperated surgical robotic arm using ESP32 microcontrollers, servo motors, wireless communication, and live video streaming. The system will enable an operator to remotely control a robotic arm while observing its operation through a real-time video feed.

2. Objectives

General Objective

To design and develop a teleoperated robotic arm with real-time vision feedback for demonstrating the fundamental principles of robotic-assisted surgery.

Specific Objectives

1. Design and construct a multi-degree-of-freedom robotic arm.
2. Develop a remote control system using potentiometers and ESP32 microcontrollers.
3. Implement wireless communication between the operator station and robotic arm.
4. Integrate a live video streaming system for visual feedback.
5. Evaluate system performance in terms of responsiveness, accuracy, and reliability.
6. Demonstrate remote manipulation of objects using teleoperation techniques.

3. Methodology

Phase 1: System Design

- Design the mechanical structure of the robotic arm.
- Select suitable servo motors and control electronics.
- Develop the system architecture.

Phase 2: Hardware Development

- Assemble the robotic arm.

- Install servo motors for joint actuation.
- Interface potentiometers with ESP32 for user control.
- Configure power management circuits.

Phase 3: Communication System

- Establish wireless communication using MQTT protocol or ESP-NOW.
- Implement command transmission between controller and robotic arm.
- Optimize data transfer for low-latency operation.

Phase 4: Real-Time Vision Feedback

- Integrate a camera system.
- Develop live video streaming between operator and robotic arm.
- Enable real-time monitoring of robotic arm movements.

Phase 5: Software Development

- Program ESP32 microcontrollers.
- Implement servo control algorithms.
- Develop data filtering and smoothing techniques.
- Create monitoring and control interfaces.

Phase 6: Testing and Evaluation

- Measure communication latency.
- Evaluate motion accuracy.
- Test real-time video performance.
- Conduct object manipulation experiments.

4. Expected Outcomes

- Functional teleoperated robotic arm prototype.
- Wireless remote control system.
- Real-time vision feedback capability.
- Demonstration of remote object manipulation.
- Low-cost educational platform for surgical robotics research.

5. Applications

- Robotic-assisted surgery research.
- Medical robotics education.
- Telemedicine and remote healthcare.
- Human-machine interaction studies.
- Robotics and automation research.

6. Conclusion

This project proposes the development of a low-cost teleoperated surgical robotic arm with real-time vision feedback. The system will demonstrate essential concepts used in modern robotic surgery, including remote manipulation, wireless communication, and visual feedback. Although designed as an educational prototype rather than a clinical device, the project provides valuable experience in robotics, embedded systems, IoT, control systems, and healthcare technology.

Recent Research Papers (2024–2026)

These papers are highly relevant to your project topic and can be included in your literature review:

1. **A Low-Cost Teleoperable Surgical Robot with a Macro-Micro Structure and a Continuum Tip for Open-Source Research (2024)** — Focuses on affordable teleoperated surgical robotics for research. ([ResearchGate](#))
2. **Teleoperation in Robot-Assisted MIS with Adaptive RCM via Admittance Control (2024)** — Presents a teleoperation framework for minimally invasive surgery. ([Springer](#))
3. **Ensuring Communication Redundancy and Establishing a Telementoring System for Robotic Telesurgery (2024)** — Discusses reliable communication architectures for telesurgery. ([Springer](#))
4. **AI Solutions for Overcoming Delays in Telesurgery and Telementoring (2024)** — Reviews AI methods for reducing latency in remote surgery. ([Springer](#))
5. **AI-Augmented Autonomous Telesurgery: Defining the Next Frontier in Intelligent Remote Surgery (2025)** — Explores future AI-assisted telesurgical systems. ([Springer](#))
6. **A Predictive Approach for Enhancing Accuracy in Remote Robotic Surgery Using Informer Model (2025)** — Addresses network delays and packet loss in remote robotic surgery. ([MDPI](#))
7. **RoboTwin: A Robotic Teleoperation Framework Using Digital Twins (2025)** — Uses digital twins to improve teleoperation performance and reduce latency. ([arXiv](#))
8. **A Vision-Based Shared-Control Teleoperation Scheme for Controlling the Robotic Arm of a Four-Legged Robot (2025)** — Demonstrates vision-assisted teleoperation of robotic manipulators. ([arXiv](#))
9. **FPGA-Based Low-Latency Semantic Feedback System for Real-Time Instrument Localization in Telesurgery (2026)** — Focuses on reducing visual latency and improving telesurgery feedback. ([Wiley Online Library](#))
10. **Internet of Robotic Things (IoRT): Architecture, Components and Challenges** — Useful for explaining the IoT communication layer of your ESP32-based robotic arm. ([arXiv](#))